Chapter 45: The Matrix of a Linear Transformation

Goal: If $T : V \to W$ is a linear transformation of an n-dimensional vector space V to an m-dimensional vector space W, and if we choose bases for V and W, then we can associate a unique $m \times n$ matrix A with T that will enable us to find $T(\mathbf{x})$ for $\mathbf{x}$ in V by doing matrix multiplication.

Recall: The **coordinate vector** of $\mathbf{v}$ with respect to B is

$$C_B(\mathbf{v}) = C_B(a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n) = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Note: $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the ith standard basis vector for $\mathbb{R}^n$. This is true for ANY V and B.

Example: Let $B = (x+1, 3)$ be an ordered basis for P_1. Find $C_B(\mathbf{v})$ for $\mathbf{v} = 2x - 4$.

$$2x - 4 = a_1 \cdot (x+1) + a_2 \cdot 3$$
$$= a_1 x + (a_1 + 3a_2)$$
$$\Rightarrow \begin{cases} a_1 = 2 \\ a_2 = -2 \end{cases} \Rightarrow C_B(v) = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

Theorem: If V has dimension n and $B = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is any ordered basis of V, the coordinate transformation $C_B : V \to \mathbb{R}^n$ is an isomorphism. In fact, $C_B^{-1} : \mathbb{R}^n \to V$ is given by

$$C_B^{-1}\left(\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}\right) = a_1 v_1 + \cdots + a_n v_n \qquad \text{for all } \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \text{ in } \mathbb{R}^n.$$

Example: Let $B = (x+1, 3)$ be an ordered basis for P_1. What's $C_B^{-1}\left(\begin{bmatrix} 4 \\ -5 \end{bmatrix}\right)$?

$$C_B^{-1}\left(\begin{bmatrix} 4 \\ -5 \end{bmatrix}\right) = 4 \cdot (x+1) - 5 \cdot 3$$
$$= 4x - 11$$

Matrix of a Linear Transformation

Let V, W be vector spaces with $\dim(V) = n$, $\dim(W) = m$

If we want to build a matrix A that does the same thing as a linear transformation $T : V \to W$, using ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ and $D = (\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m)$ for V and W, what would we have to do?

To compute $T(\mathbf{v})$ using the matrix A:

1. Build the coord. transf. coordinate of $\mathbf{v}$ with respect to B, $C_B(\mathbf{v})$.
2. Multiply $C_B(\mathbf{v})$ by A to make $AC_B(\mathbf{v}) = T_A(C_B(\mathbf{v}))$.
3. "Translate" $AC_B(\mathbf{v})$ back to W: $C_D(T(v)) = A \cdot C_B(v)$

The following diagram illustrates the relationship between T and A:

$$\begin{tikzcd} V \ni \cdot v \arrow[r, "T"] \arrow[d, shift right, "C_B"'] & W \arrow[d, shift left, "C_D"] \\ \mathbb{R}^n \ni \cdot C_B(v) \arrow[r, "T_A"] \arrow[u, shift right, "C_B^{-1}"'] & \mathbb{R}^m \arrow[u, shift left, "C_D^{-1}"] \end{tikzcd}$$

$$T : V \to W$$

$$T_A : \mathbb{R}^n \to \mathbb{R}^m$$

If we compute $\mathbf{b}_j$ using both sides, we obtain the following information about A:

$$T(v) = C_D^{-1} \cdot A \cdot C_B(v) \qquad T_A(C_B(v)) = A \cdot C_B(v)$$

Theorem: Let $T : V \to W$ be a linear transformation of an n-dimensional vector space V into an m-dimensional vector space W, and let $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ and $D = (\mathbf{d}_1, \ldots, \mathbf{d}_m)$ be ordered bases for V and W, respectively. Then the $m \times n$ matrix A whose jth column is $C_D[T(\mathbf{b}_j)]$ has the following property:

$$C_D[T(\mathbf{v})] = AC_B(\mathbf{v}) \text{ for every } \mathbf{v} \text{ in } V.$$

Moreover, A is the only matrix with this property.

The matrix A is called the **matrix of T corresponding to the ordered bases** B and D, and we use the following notation:

$$M_{DB}(T) = \begin{bmatrix} C_D[T(\mathbf{b}_1)] & C_D[T(\mathbf{b}_2)] & \cdots & C_D[T(\mathbf{b}_n)] \end{bmatrix}$$

Note: We'll denote this matrix as $M_{DB}(T)$ (instead of A) from now on. Your book denotes it as $[T]_{DB}$.

Example: Let $L : P_2 \to P_1$ be defined by $L(p) = p'$ and consider the ordered bases $B = (x^2, x, 1)$ and $D = (x, 1)$ for P_2 and P_1, respectively.

(a) Find the matrix $M_{DB}(L)$ of L corresponding to B and D.

(b) For $p = x^2 - 5x + 7$, compute $L(p)$ both directly and by using $M_{DB}(L)$.

(a) $C_D(L(\vec{b}_1)) = C_D(L(x^2)) = C_D(2x) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$C_D(L(\vec{b}_2)) = C_D(1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$C_D(L(\vec{b}_3)) = C_D(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow M_{DB}(L) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix}$

(b) $L(p) = 2x - 5$, $L(p) = C_D^{-1} \cdot M_{DB}(L) \cdot C_B(p) = 2x - 5$.

Example: Let L be the same linear function as above, but use ordered bases $\widetilde{B} = (1, x, x^2)$ and $\widetilde{D} = (x+1, x-1)$ for P_2 and P_1 respectively.

(a) Find the matrix $M_{\widetilde{D}\widetilde{B}}(L)$ of L corresponding to $\widetilde{B}$ and $\widetilde{D}$.

(b) If $p = x^2 - 5x + 7$, compute $L(p)$ by using $M_{\widetilde{D}\widetilde{B}}(L)$.

(a) $M_{\widetilde{D}\widetilde{B}}(L) = \left[C_D(L(\tilde{b}_1)), \ C_D(L(\tilde{b}_2)), \ C_D(L(\tilde{b}_3)) \right]$

$= \begin{bmatrix} 0 & \frac{1}{2} & 1 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix}$

(b). $L(p) = C_D^{-1} \cdot M_{\widetilde{D}\widetilde{B}}(L) \cdot C_B(p) = C_D^{-1}\left(\begin{bmatrix} 0 & \frac{1}{2} & 1 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ -5 \\ 1 \end{bmatrix} \right)$

$= C_D^{-1}\left(\begin{bmatrix} -\frac{3}{2} \\ \frac{7}{2} \end{bmatrix} \right)$

$= -\frac{3}{2}(x+1) + \frac{7}{2}(x-1)$

$= 2x - 5.$

We will now see that composition of linear transformations is equivalent to matrix multiplication:

Theorem: Let $T : V \to W$, and $S : W \to U$ be linear transformations and let B, D, and E be finite ordered bases of V, W, and U, respectively. Then

$$M_{EB}(ST) = M_{ED}(S) \cdot M_{DB}(T)$$

Properties of a transformation from properties of its matrix

Theorem: Let $T : V \to W$ be a linear transformation, with ordered bases B and D of V and W, respectively. Then:

1. A vector $\mathbf{v}$ in V is in $\ker T$ if and only if $C_B(\mathbf{v})$ is in the null space of $M_{DB}(T)$.
2. The dimension of the kernel of T is equal to dimension of the null space of $M_{DB}(T)$.
3. A vector $\mathbf{w}$ in W is in $\operatorname{im} T$ if and only if $C_D(\mathbf{w})$ is in the column space of $M_{DB}(T)$.
4. The dimension of the image of T is equal to the dimension of the column space of $M_{DB}(T)$.

Proof: (Pick 1 or 3):

1. "$\Rightarrow$" Assume $v \in \ker T$, then $T(v) = \vec{0}_W$

$\Rightarrow \quad M_{DB}(T) \cdot C_B(v) = C_D(T(v))$

$\Rightarrow \quad M_{DB}(T) \cdot C_B(v) = C_D(\vec{0}_W) = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^m.$

$\Rightarrow C_B(v) \in Null(M_{DB}(T))$

"$\Leftarrow$" Assume $C_B(v) \in Null(M_{DB}(T))$, then

$C_D^{-1}(M_{DB}(T) \cdot C_B(v)) = T(v)$

$\Rightarrow C_D^{-1}\left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}\right) = T(v) \Rightarrow T(v) = \vec{0}_W$. So $v \in \ker T$.

Theorem: Let $T : V \to W$ be a linear transformation, where $\dim V = \dim W = n$. The following are equivalent.

1. T is an isomorphism.
2. For every choice of ordered bases B and D of V and W, $M_{DB}(T)$ is invertible.
3. For at least one choice of ordered bases B and D of V and W, $M_{DB}(T)$ is invertible.

When this is the case, $[M_{DB}(T)]^{-1} = M_{BD}(T^{-1})$.

Example: $L : P_2 \to P_1$ be defined by $L(p) = p'$. Compute $\dim \operatorname{im} L$.

$\dim(\operatorname{Im} L) = \dim(\operatorname{col}(M_{DB}(L))) = 2$.

Example: Let $T : P_1 \to P_1$ by $T(p) = xp'$. Is T an isomorphism?

NO. $T(x+1) = x = T(x)$

NOT 1-1.

Chapter 46: Similarity of Linear Operators

Recall: A linear transformation from V to itself i.e. $T : V \to V$ is called a **linear operator**.

In this case, we can choose the SAME ordered basis B for V and the target space (also V). The matrix of T in this case has a special name/notation:

Definition: If $T : V \to V$ is an operator on a vector space V, and if B is an ordered basis of V, define $M_B(T) = M_{BB}(T)$. Some books call this the ***B*-matrix** of T. Your book writes $M_B(T)$ as $[T]_B$.

Question: For a linear operator $T : V \to V$, how does $M_B(T)$ change depending on the choice of basis B?

Starting question: How do the coordinates of a vector change when we change bases?

Recall: For a vector space V with ordered bases $B = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ and D, the transition matrix $P_{D\leftarrow B} = \begin{bmatrix} C_D(\mathbf{v}_1) & C_D(\mathbf{v}_2) & \cdots & C_D(\mathbf{v}_n) \end{bmatrix}$ gives us

$$P_{D\leftarrow B} C_B(\mathbf{v}) = C_D(v)$$

Example: Let $B = (1, x)$ and $D = (x+1, x-1)$ be two ordered bases for P_1, and build $P_{D\leftarrow B}$.

$$1 = \frac{1}{2}(x+1) - \frac{1}{2}(x-1)$$

$$x = \frac{1}{2}(x+1) + \frac{1}{2}(x-1)$$

$$P_{D\leftarrow B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

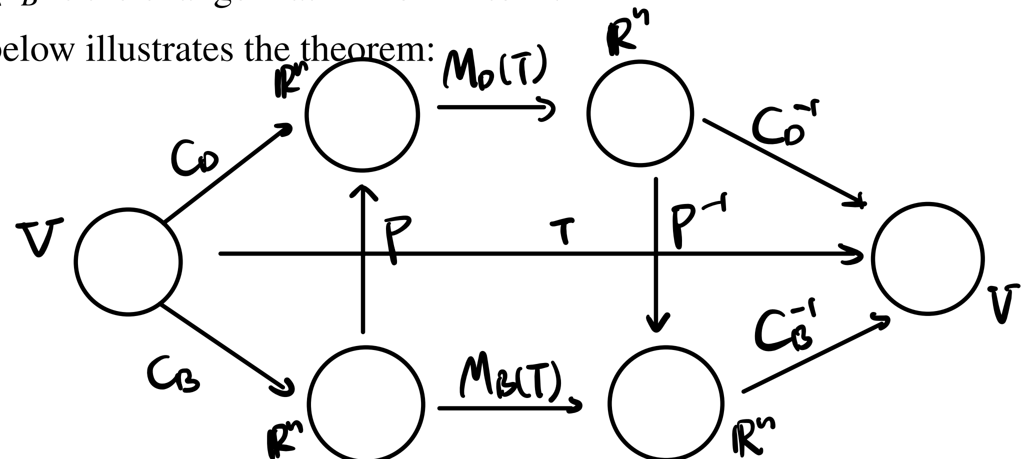
$$P^{-1}_{D\leftarrow B} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Theorem: For ordered bases B and D for a vector space V, $P_{D\leftarrow B} = M_{DB}(T)$ for $T = I_V$

Set-up: Suppose we have a linear operator $T : V \to V$, and two ordered bases B and D for V. How do $M_B(T)$ and $M_D(T)$ relate?

Theorem: (Similarity Theorem) Let B and D be two ordered bases of a finite dimensional vector space V. If $T : V \to V$ is any linear operator, the matrices $M_B(T)$ and $M_D(T)$ of T with respect to these bases are similar. More precisely, $(v_1 \cdots v_n) = (u_1 \cdots u_n) P$
$M_B(T) = P^{-1} M_D(T) P =$
where $P = P_{D \leftarrow B}$ is the change matrix from B to D.

The diagram below illustrates the theorem:



Example: Let $L : P_1 \to P_1$ by $L(ax+b) = bx + (2a+b)$. Using ordered bases $B = (1, x)$ and $D = (x+1, x-1)$:
(a) Build $M_D(L)$.
(b) Use the Similarity Theorem to compute $M_B(L)$.

(a).
$$M_D(L) = [\, C_D(L(d_1)) \;\; C_D(L(d_2)) \,]$$
$$= [\, C_D(x+3) \;\; C_D(-x+1) \,]$$
$$= \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix}$$

(b)
$$M_B(L) = P_{D\leftarrow B}^{-1} M_D(L) P_{D\leftarrow B}$$
$$= \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$
$$= [\, C_B(L(b_1)) \;\; C_B(L(b_2)) \,].$$

Example: Let $L : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$L\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} 2u_1 - u_3 \\ u_1 + u_2 - u_3 \\ u_3 \end{bmatrix}.$$

Let D be the standard basis for $\mathbb{R}^3$. Then the D-Matrix of L is

$D = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$M_D(L) = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now consider the ordered basis $B = \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)$ for $\mathbb{R}^3$. What is $M_B(L)$, given that the transition matrix P from B to D has inverse

$$P^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}?$$

$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$

$$M_B(L) = P^{-1} M_D(L) \cdot P$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}}$$

This raises the question: Given $T : V \to V$, can we always find a basis B such that the B-matrix of T is diagonal? If not, when can we find such a basis?

Eigenvalues of Linear Operators

Definition: A real number λ is called an **eigenvalue** of an operator $T : V \to V$ if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$ in V. In this case, $\mathbf{v}$ is called an **eigenvector** of T corresponding to λ.

Example: Consider $T : P_2 \to P_2$ by $T(p(x)) = xp'(x)$. Show that x^2 is an eigenvector of T. What's the eigenvalue?

$$T(x^2) = x \cdot (x^2)' = 2x^2$$

$$\lambda = 2.$$

The subspace $E_\lambda(T) = \{\mathbf{v} \text{ in } V | T(\mathbf{v}) = \lambda\mathbf{v}\}$ is called the **eigenspace** of T corresponding to λ.

Remark: If A is an $n \times n$ matrix, then λ is an eigenvalue of the matrix operator $T_A : \mathbb{R}^n \to \mathbb{R}^n$ if and only if λ is an eigenvalue of the matrix A, and the eigenspaces are the same.

How do we find eigenvalues/eigenvectors of a general linear operator?

Choose an ordered basis B for V, and consider $A = M_B(T)$. Suppose $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$.

What is $AC_B(\mathbf{v})$?

$$AC_B(v) = C_B(T(v))$$

$$= C_B(\lambda v)$$

$$= \lambda C_B(v)$$

Theorem: Suppose $T : V \to V$ is a linear operator where $\dim V = n$, λ is a number, and $\mathbf{v}$ is a vector in V. Then the following are equivalent:

1. T has eigenvalue/eigenvector pair $\lambda, \mathbf{v}$.
2. For every choice of ordered basis B for V, the matrix $M_B(T)$ has eigenvalue/eigenvector pair $\lambda, C_B(\mathbf{v})$.
3. For at least one choice of ordered basis B for V, the matrix $M_B(T)$ has eigenvalue/eigenvector pair $\lambda, C_B(\mathbf{v})$.

This theorem says: To find the eigenvalues/eigenvectors for a general linear operator $T: V \to V$:

1. Pick any ordered basis B for V.
2. Build the matrix $M_B(T)$.
3. Compute the eigenvalues and eigenvectors for $M_B(T)$.
4. T has the same eigenvalues. To get the eigenvectors, "translate" back to V from $\mathbb{R}^n$ using C_B^{-1}.

Example: Let $L: P_1 \to P_1$ by $L(ax+b) = bx + (2a+b)$. We computed $M_B(L)$ and $M_D(L)$ for both ordered bases $B = (1, x)$ and $D = (x+1, x-1)$:

$$M_B(L) = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \qquad M_D(L) = \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix}$$

Find the eigenvalues and corresponding eigenspaces for each matrix, then find the eigenvalues and eigenspaces for L.

$$\det(\lambda I - M_B(L)) = \begin{vmatrix} \lambda-1 & -2 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - \lambda - 2$$
$$= (\lambda-2)(\lambda+1)$$

$\lambda_1 = 2$, $\lambda_2 = -1$

- For $\lambda_1 = 2$, $(2I - M_B(L)) \cdot \vec{v} = \vec{0} \Rightarrow \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

  $E_2(M_B(L)) = \operatorname{span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$

- For $\lambda_2 = -1$, $(-I - M_B(L)) \cdot \vec{v} = \vec{0} \Rightarrow \vec{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

  $E_{-1}(M_B(L)) = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$.

L also has eigen-values $2, -1$

$E_2(L) = \operatorname{span}\{x+2\}$, $E_{-1} = \operatorname{span}\{-x+1\}$.

(more room for example)

$D = \{1, x, x^2\}$.

Example: Let $T: P_2 \to P_2$ be a linear operator defined by

$$T(ax^2+bx+c) = -bx-2c.$$

What are the eigenvalues of T?

$$M_D(T) = [C_D(T(1)), C_D(T(x)), C_D(T(x^2))]$$

$$= [C_D(-2), C_D(-x), C_D(0)]$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Diagonalizability

Recall: An $n \times n$ matrix is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ consisting of eigenvectors of A.

Definition: A linear operator T on a finite dimensional space V is called **diagonalizable** if there is an ordered basis B of V such that $M_B(T)$ is diagonal.

Theorem: Let $T: V \to V$ be a linear operator on a finite dimensional space V. Then the following conditions are equivalent.

1. V has a basis consisting of eigenvectors of T.
2. There exists a basis B of V such that $M_B(T)$ is diagonal.

Definition: A linear operator T on a finite dimensional space V is called diagonalizable if V has a basis consisting of eigenvectors of T.

Example: Let $L : P_1 \to P_1$ by $L(ax+b) = bx + (2a+b)$. Find a basis U of P_1 such that $M_U(L)$ is diagonal.

$B = \{x+2, -x+1\}$.

Corollary: Let $T : V \to V$ be a linear operator on a finite dimensional space V. T is diagonalizable if and only if a matrix $M_B(T)$ representing T is diagonalizable.

Example: Let $T : P_2 \to P_2$ be a linear operator defined by $T(ax^2+bx+c) = -bx-2c$. Is T diagonalizable?

Yes, b/c $M_D(L) = \begin{bmatrix} -2 & & \\ & -1 & \\ & & 0 \end{bmatrix}$

is a diagonal matrix.